

Linux Cheat Sheet Bundle

Cybersecurity & IT Edition — Beginner to Intermediate

This bundle is designed for Cybersecurity students, Linux beginners, IT support learners, and self-taught developers. It combines practical Linux usage with real-world cybersecurity workflows.

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1. File Navigation

Command	Description	Example
pwd	Print current directory	pwd
ls -la	List hidden files with details	ls -la /var/log
cd	Change directory	cd /etc
tree	Display folder structure	tree ~/projects
mkdir	Create directories	mkdir recon_reports
rm -rf	Delete recursively	rm -rf old_logs
cp -r	Copy files/folders	cp -r backup/ archive/
mv	Move or rename	mv report.txt report_old.txt
touch	Create file	touch notes.txt
cat	View file contents	cat auth.log

Absolute vs Relative Paths

Absolute paths start from the root directory (/). Relative paths depend on your current directory. Examples: /home/kondwani/projects (absolute) ../projects (relative)

Wildcard Usage

```
*.txt      # All text files
?.log     # Single-character filename match
```

```
rm *.tmp    # Remove temp files
```

nano & vim Basics

```
nano notes.txt
CTRL + O  -> Save
CTRL + X  -> Exit
```

```
vim notes.txt
i          -> Insert mode
ESC        -> Command mode
:wq        -> Save & Quit
:q!        -> Force Quit
```

Quick Tips

- Use TAB for autocomplete.
- Avoid using `rm -rf` as root unless necessary.
- Use `tree` for faster project visualization.

Practice Exercises

1. Create a directory called `pentest_lab`.
2. Create 3 files inside it.
3. Move one file into a subdirectory.
4. Display the directory structure using `tree`.

2. Permissions & Ownership

Linux permissions determine who can read, write, or execute files.

Permission	Meaning
r	Read
w	Write
x	Execute

chmod

```
chmod 755 script.sh
chmod u+x tool.py
```

755 = Owner: rwx | Group: r-x | Others: r-x

644 = Owner: rw- | Group: r-- | Others: r--

Special Permissions

```
chmod +t shared_folder  # Sticky bit
chmod u+s binary_file    # SUID
chmod g+s project_dir    # SGID
```

Security-Focused Example

```
chmod 600 id_rsa
```

Restricts private SSH keys to the owner only.

Common Mistakes

- Setting files to 777 unnecessarily.
- Forgetting executable permissions on scripts.
- Running sudo commands blindly.

Practice Exercises

1. Create a script and make it executable.
2. Change ownership of a directory.
3. Experiment with octal permissions.

3. grep / find / Searching

Command	Purpose	Example
grep	Search text	grep 'error' logs.txt
find	Find files	find / -name '*.conf'
locate	Fast indexed search	locate apache2.conf
which	Find executable path	which python3
whereis	Locate binaries/docs	whereis ssh
xargs	Pass output as arguments	cat ips.txt xargs ping

Regex Examples

```
grep '^root' /etc/passwd
grep '[0-9]\{3\}' logs.txt
```

Cybersecurity Use Cases

```
# Find failed SSH logins
grep "Failed password" /var/log/auth.log

# Find SUID binaries
find / -perm -4000 2>/dev/null

# Find large suspicious files
find / -size +100M
```

Piping with grep

```
ps aux | grep apache
netstat -tulnp | grep 22
```

Practice Exercises

1. Search a log file for failed login attempts.
2. Find all .conf files under /etc.
3. Search for running Python processes.

4. Networking Commands

Command	Purpose	Example
ping	Connectivity test	ping google.com
ip addr	Show interfaces	ip addr show
ss -tulnp	Socket statistics	ss -tulnp
traceroute	Trace network path	traceroute 8.8.8.8
curl	Transfer data	curl ifconfig.me
wget	Download files	wget file.zip
dig	DNS lookup	dig openai.com
nmap	Port scanning	nmap -sV 192.168.1.1
ssh	Remote access	ssh user@host
scp	Secure copy	scp file.txt user@host:/tmp

TCP vs UDP

TCP is connection-oriented and reliable.

UDP is faster but does not guarantee delivery.

Port Concepts

Ports identify services running on a system.

Example: 22 = SSH

80 = HTTP

443 = HTTPS

Troubleshooting Workflow

1. ping target
2. ip addr
3. ss -tulnp
4. traceroute target
5. dig domain.com

Practice Exercises

1. Scan your local subnet with nmap.
2. Use curl to fetch a webpage header.
3. Identify listening ports on your machine.

5. Package Management

Debian / Ubuntu

```
sudo apt update
sudo apt install nmap
sudo apt remove apache2
dpkg -l
```

RedHat / CentOS

```
sudo dnf install wireshark
sudo yum update
rpm -qa
```

Arch Linux

```
sudo pacman -Syu
sudo pacman -S neofetch
```

Security Tips

- Install packages only from trusted repositories.
- Verify package signatures when possible.
- Avoid random install scripts from unknown GitHub repos.

Package Manager Differences

Manager	Distribution
apt	Debian/Ubuntu
yum/dnf	RedHat/CentOS/Fedora
pacman	Arch Linux

Practice Exercises

1. Install and remove a package.
2. Search for a package using apt search.
3. Update all installed packages safely.

6. Process Monitoring & System Monitoring

Command	Purpose
ps aux	List processes
top	Live process monitor
htop	Interactive monitor
kill	Terminate process
jobs	Show background jobs
systemctl	Manage services
journalctl	View system logs
df -h	Disk usage
free -m	Memory usage
uptime	System uptime

Suspicious Process Hunting

```
ps aux --sort=-%mem
top
```

```
lsof -i
```

Zombie Processes

Zombie processes are terminated processes whose parent has not yet read their exit status.

Linux Service Basics

```
systemctl status ssh
systemctl restart apache2
journalctl -xe
```

Practice Exercises

1. Kill a test process.
2. Monitor memory usage in real time.
3. Restart a system service safely.

7. Linux for Cybersecurity

Linux is foundational in Cybersecurity because most servers, cloud systems, and security tools run on Linux-based environments.

Area	Linux Relevance
Penetration Testing	Kali Linux tools
SOC Analysis	Log analysis and automation
Cloud Security	Linux server administration
Malware Analysis	Sandboxing and reverse engineering
Digital Forensics	File system and memory inspection

Core Skill Stack

- Shell scripting
- Networking
- Permissions management
- Process monitoring
- Automation with Bash/Python

8. Condensed One-Page Cheat Sheet

Category	Command
Navigation	ls -la / cd / pwd
Permissions	chmod 755 file
Search	grep 'error' log.txt
Networking	ss -tulnp
Packages	sudo apt update
Processes	ps aux / top
Logs	journalctl -xe

Cybersecurity Quick Workflow:

1. Enumerate system
2. Inspect processes
3. Review logs
4. Verify network connections
5. Check permissions
6. Audit services

End of Bundle — Keep Practicing. Linux skill comes from repetition and real usage.